

Overview of Matrices

ECON 301

Spring 2022



What is a matrix?

↳ A square (rectangular) array of numbers

↓
Two dimensions

- # of rows (m)
- # of columns (n)

Example: If A is $m \times n$ matrix $\rightarrow A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$

↳ The element in the
 i -th row, j -th column is a_{ij}

$$B = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 1 & 6 & 0 & 1 \end{bmatrix} \quad B \text{ is } 2 \times 4 \quad \begin{matrix} b_{21} = 1 \\ b_{23} = 0 \end{matrix}$$

A shorthand notation $A = [a_{ij}]_{m \times n}$

A square matrix is one with the same # of rows and columns

$$S = [s_{ij}]_{m \times m}, \quad \text{if } m = 4 \quad S = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 1 & 1 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

A vector is also a matrix:

• row vectors is a $1 \times m$ matrix: $(x_1, x_2, \dots, x_m)$

• column vectors is a $n \times 1$ matrix: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

Diagonal Matrix

A is a diagonal (square) matrix if $a_{ij} = 0$ for all $i \neq j$ and $a_{ij} \neq 0$ for $i = j$

$$A = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & & 0 \\ & & \ddots & \\ 0 & & & a_{mm} \end{bmatrix}$$

Identity Matrix and the Zero Matrix

The $n \times n$ Identity matrix

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & 0 \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix}$$

The $m \times n$ Zero matrix

$$O = \begin{bmatrix} 0 & 0 & \dots & 0 \\ & & \ddots & \\ 0 & & & 0 \end{bmatrix}$$

$$3x_1 + 4x_2 - x_3 = 0$$

$$2x_1 + 5x_2 + 2x_3 = 10$$

$$x_1 - x_2 + 3x_3 = 5$$

System of 3 linear eqs. in 3 unknowns



$$Ax = b$$

$$A = \begin{bmatrix} 3 & 4 & -1 \\ 2 & 5 & 2 \\ 1 & -1 & 3 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 10 \\ 5 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Operations with matrices

- Matrix addition: A and B two $m \times n$ matrices (same size)

$$A = [a_{ij}] \quad B = [b_{ij}] \quad A + B = [a_{ij} + b_{ij}]$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$$

- Scalar Multiplication: If γ is a scalar ^{real number} and $A = [a_{ij}]$ is an $m \times n$ matrix

$$\gamma A = [\gamma a_{ij}]$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad \gamma A = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}$$

$$\gamma = 2$$

Matrix multiplication

If A and B are two matrices, to form

AB then the # of columns of A must equal the # of rows of B

↓
dimensions must be
congruent!

A is $m \times n$
 B is $n \times p$ } AB

$$\begin{array}{c} \text{A} \\ \left[\begin{array}{cccc} a_{i1} & a_{i2} & \dots & a_{in} \end{array} \right] \end{array} \begin{array}{c} \text{B} \\ \left[\begin{array}{c} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{array} \right] \end{array} = \begin{array}{c} \text{AB} \\ \left[\begin{array}{c} \sum_{k=1}^n a_{ik} b_{kj} \end{array} \right] \end{array}$$

i -th row of A

j -th column of B

(i,j) -th component of AB

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 5 \end{bmatrix} \rightarrow 2 \times 3$$

$$B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 3 & 0 \end{bmatrix} \rightarrow 3 \times 3$$

$$AB = \begin{bmatrix} 1 \cdot 0 + 3 \cdot 1 + 2 \cdot 1 & 1 \cdot 2 + 3 \cdot 4 + 2 \cdot 3 & 1 \cdot 1 + 3 \cdot 3 + 2 \cdot 0 \\ 4 \cdot 0 + 0 \cdot 1 + 5 \cdot 1 & 4 \cdot 2 + 0 \cdot 4 + 5 \cdot 3 & 4 \cdot 1 + 0 \cdot 3 + 5 \cdot 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 20 & 10 \\ 5 & 23 & 4 \end{bmatrix}$$

$$(2 \times 3) \times (3 \times 3) = 2 \times 3$$

Matrix multiplication is NOT commutative.

↳ They might not be congruent, like the example above: AB can be formed but BA cannot

↳ Even if AB and BA can be formed, in general $AB \neq BA$

Inverse (Only exist for square matrices)

Suppose A is a $n \times n$ square matrix

A has an inverse if there exists a matrix B

$$\text{such that } AB = I_n$$

$$BA = I_n$$

If B exists A is called non-singular and B is denoted by A^{-1}

$$A_{n \times n} I_n = A$$

$$I_n \cdot A_{n \times n} = A$$